Nonlocal corrections to localized surface plasmons

This set of notes follows the papers, Nonlocal Response of Metallic Nanospheres Probed by Light, Electrons, and Atoms (https://pubs.acs.org/doi/10.1021/nn406153k) and Spatial Nonlocality in the Optical Response of Metal Nanoparticles (https://pubs.acs.org/doi/10.1021/jp204261u).

The article expands the electric field as $\mathbf{E} = \sum_\nu a_\nu \mathbf{M}_\nu + b_\nu \mathbf{N}_\nu + c_\nu \mathbf{L}_\nu$

Note that $\mathbf{M}_\nu = \mathbf{L}\psi_\nu, \mathbf{N}_\nu = (1/k)\nabla \times \mathbf{M}_\nu$ The latter quantity is trivially transverse (divergence-free). The former is also transverse because

$$\nabla \cdot (\mathbf{L}\psi) \propto \nabla \cdot (\mathbf{r} \times \nabla\psi) = \partial_i \varepsilon_{ijk}(r_j \partial_k \psi) = \varepsilon_{ijk} r_j \partial_i \partial_k \psi = \varepsilon_{ijk} r_j \partial_k \partial_i \psi = -\varepsilon_{kji} r_j \partial_k \partial_i \psi = -\varepsilon_{ijk} r_j \partial_i \partial_k \psi$$

In contrast, the last quantity, $\mathbf{L}_\nu = (1/k)\nabla\psi_\nu$ is clearly longitudinal with no transverse component.

Notational point

The paper by Christensen et al introduces the quantity, k, above It's not entirely clear to me whether this is the free space transverse wavevector, k_0 or what. In the below, we set $k \equiv k_0$. Ultimately, this makes no difference in calculating the scattering coefficients. All other notation used below should (hopefully) closely follow that of the Christensen paper.

Derivations of equations 1(a, b) and 10(a, b) (of Christensen paper)

We start with Maxwell's equations:

$$\nabla \times \mathbf{E}(\mathbf{r}, \omega) = i\omega\mu_0 \mathbf{H}(\mathbf{r}, \omega), \nabla \times \mathbf{H}(\mathbf{r}, \omega) = -i\omega\varepsilon_0\varepsilon_\infty \mathbf{E}(\mathbf{r}, \omega) + \mathbf{J}(\mathbf{r}, \omega)$$

where the high frequency contribution to the polarization is subsumed into a permittivity, ε_∞, and the free current density, which is sourced by the electron liquid, and for which we write an equation of motion below, is given by $\mathbf{J}(\mathbf{r}, \omega)$. We have, upon combining the two equations, the following:

$$\nabla \times \nabla \times \mathbf{E}(\mathbf{r},\omega) = i\omega\mu_0(-i\omega\varepsilon_0\varepsilon_\infty\mathbf{E}(\mathbf{r},\omega) + \mathbf{J}(\mathbf{r},\omega)) = k_0^2\varepsilon_\infty\mathbf{E}(\mathbf{r},\omega) + i\omega\mu_0\mathbf{J}(\mathbf{r},\omega),$$

which is **Equation 1b** of the Christensen paper. In the hydrodynamic model, the current density associated with the electron liquid is related to the total electric field via a modified current-field relation:

$$\frac{\beta_F^2}{\omega(\omega+i\eta)}\nabla(\nabla\cdot\mathbf{J}(\mathbf{r},\omega)) + \mathbf{J}(\mathbf{r},\omega) = \sigma(\omega)\mathbf{E}(\mathbf{r},\omega),$$

which is **Equation 1a** of the first paper. We may alternatively write this, using the conservation law, $\nabla\cdot\mathbf{J}(\mathbf{r},\omega) = i\omega\rho(\mathbf{r},\omega)$, as well as the local conductivity $\sigma(\omega) = i\omega_p^2\varepsilon_0/(\omega+i\eta)$ in the following way:

$$\beta_F^2\nabla\rho(\mathbf{r},\omega) - i(\omega+i\eta)\mathbf{J}(\mathbf{r},\omega) = \omega_p^2\varepsilon_0\mathbf{E}(\mathbf{r},\omega)$$

We now seek to prove **Equation 10a** of the paper, which gives an equation of motion for the transverse fields. To proceed, we first note that by taking the curl of the current-field relation, we obtain the useful relation:

$$\nabla\times\mathbf{J}(\mathbf{r},\omega) = \sigma(\omega)\nabla\times\mathbf{E}(\mathbf{r},\omega)$$

Therefore, by taking the curl of our previous equation for the quantity $\nabla\times\nabla\times\mathbf{E}(\mathbf{r},\omega)$ we obtain:

$$\nabla\times\nabla\times(\nabla\times\mathbf{E}(\mathbf{r},\omega)) = [k_0^2\varepsilon_\infty + i\omega\mu_0\sigma(\omega)]\nabla\times\mathbf{E}(\mathbf{r},\omega)$$

We rewrite this as:

$$-\nabla^2(\nabla\times\mathbf{E}(\mathbf{r},\omega)) + \nabla(\nabla\cdot(\nabla\times\mathbf{E}(\mathbf{r},\omega))) = [k_0^2\varepsilon_\infty + i\omega\mu_0\sigma(\omega)]\nabla\times\mathbf{E}(\mathbf{r},\omega)$$

The divergence of a curl is zero, so we get:

$$\nabla^2(\nabla\times\mathbf{E}(\mathbf{r},\omega)) + [k_0^2\varepsilon_\infty + i\omega\mu_0\sigma(\omega)](\nabla\times\mathbf{E}(\mathbf{r},\omega)) = 0,$$

from which we may read off the transverse wavevector:

$$k_M^2 = k_0^2(\varepsilon_\infty + i\omega\mu_0\sigma(\omega)c^2/\omega^2) = k_0^2(\varepsilon_\infty - \sigma(\omega)/i\omega\varepsilon_0)$$

We have thus proven the relation given by **Equation 10a**. In order to obtain the equation of motion for the longitudinal fields, we proceed by taking the divergence of the current-field relation, which gives us:

$$\frac{\beta_F^2}{\omega(\omega+i\eta)}\nabla^2(\nabla\cdot\mathbf{J}(\mathbf{r},\omega))+\nabla\cdot\mathbf{J}(\mathbf{r},\omega)=\sigma(\omega)\nabla\cdot\mathbf{E}(\mathbf{r},\omega)$$

Furthermore, by taking the divergence of Ampere's law, we obtain:

$$k_0^2\varepsilon_\infty\nabla\cdot\mathbf{E}(\mathbf{r},\omega)+i\omega\mu_0\nabla\cdot\mathbf{J}(\mathbf{r},\omega)=0\rightarrow\nabla\cdot\mathbf{J}(\mathbf{r},\omega)=(ik_0^2\varepsilon_\infty/\omega\mu_0)\nabla\cdot\mathbf{E}(\mathbf{r},\omega)$$

Therefore, we have:

$$\frac{\beta_F^2}{\omega(\omega+i\eta)}\nabla^2(\nabla\cdot\mathbf{E}(\mathbf{r},\omega))+\nabla\cdot\mathbf{E}(\mathbf{r},\omega)+(i\sigma(\omega)\omega\mu_0/k_0^2\varepsilon_\infty)\nabla\cdot\mathbf{E}(\mathbf{r},\omega)=0$$

Equivalently,

$$\left[\nabla^2+\frac{\omega^2}{\beta_F^2}\left[1+\frac{i\sigma(\omega)\omega\mu_0c^2}{\omega^2\varepsilon_\infty}\right]\right]\nabla\cdot\mathbf{E}(\mathbf{r},\omega)=0$$

We simplify the longitudinal wavevector by writing,

$$k_{NL}^2\equiv\frac{\omega^2}{\beta_F^2}\left[1+\frac{i\sigma(\omega)}{\omega\varepsilon_0\varepsilon_\infty}\right]=\frac{\omega_p^2}{\beta_F^2}\times\frac{\omega^2}{\omega_p^2}\times\left[1-\frac{\omega_p^2}{\omega^2\varepsilon_\infty}\right]$$

In order to prove **Equation 10b**, we rewrite this expression for k_{NL} slightly so that we may obtain it in the form expressed in the paper. Explicitly, we note the following equalities:

$$\omega_p^2/\omega^2=\varepsilon_\infty-\varepsilon_M\rightarrow\omega^2/\omega_p^2=(\varepsilon_\infty-\varepsilon_M)^{-1}\text{ and }\left[1-\frac{\omega_p^2}{\omega^2\varepsilon_\infty}\right]=\varepsilon_M/\varepsilon_\infty$$

From these two relations, we readily obtain:

$$k_{NL}^2=\frac{\omega_p^2}{\beta_F^2}\frac{1}{\varepsilon_\infty-\varepsilon_M}\frac{\varepsilon_M}{\varepsilon_\infty},$$

which is the form of the nonlocal wavevector reported in the paper. We have thus proven the relation given by **Equation 10b**. In the case of a spherical system, since we have $(\nabla^2 + k_{NL}^2)(\nabla^2\psi) = 0$, this is satisfied if ψ is taken to be a product of a spherical harmonic and a spherical Bessel function with argument $k_{NL}r$.

Derivation of Mie parameters with nonlocality taken into account

In order to obtain the Mie scattering coefficients, we apply the three boundary conditions (continuity of tangential parts of electric and magnetic fields and vanishing of the normal component of the current at the surface), in conjunction with the modified current-field relation, which we reiterate below:

$$\beta_F^2 \nabla\rho(\mathbf{r},\omega) - i(\omega + i\eta)\mathbf{J}(\mathbf{r},\omega) = \omega_p^2\varepsilon_0\mathbf{E}(\mathbf{r},\omega)$$

Since we cannot have a normal ($\mathbf{e}_r$) component to the current at the interface, we must have (at the boundary of the sphere):

$$\beta_F^2 \partial_r\rho(\mathbf{r},\omega) = \omega_p^2\varepsilon_0\mathbf{E}(\mathbf{r},\omega)\cdot\mathbf{e}_r \text{ everywhere at the surface of the sphere}$$

If we write the electric field as $\mathbf{E}(\mathbf{r},\omega) = \frac{1}{k}\nabla\psi^L(\mathbf{r},\omega) - \frac{i}{k}\nabla\times\mathbf{L}\psi^E(\mathbf{r},\omega)$, the contribution from the first part (the longitudinal contribution) will be given by $(1/k)\mathbf{e}_r\cdot\nabla\Psi^L(\mathbf{r},\omega) = (1/k)(\partial_r\psi^L(\mathbf{r},\omega))$ and the contribution from the second part (the transverse contribution) will be:

$$\mathbf{e}_r\cdot\left[-\frac{i}{k}\nabla\times\mathbf{L}\psi^E(\mathbf{r},\omega)\right] = -(1/k)\mathbf{e}_r\cdot\nabla\times(\mathbf{r}\times\nabla\psi^E(\mathbf{r},\omega))$$

In order to evaluate this, we note that

$$\begin{aligned}\mathbf{r}\times\nabla\psi^E(\mathbf{r},\omega) = (\mathbf{e}_r\times\mathbf{e}_\theta)(r\nabla_\theta\psi^E(\mathbf{r},\omega)) + (\mathbf{e}_r\times\mathbf{e}_\phi)(r\nabla_\phi\Psi^E(\mathbf{r},\omega)) = \\ \mathbf{e}_\phi\partial_\theta\psi^E(\mathbf{r},\omega) - \frac{\mathbf{e}_\theta}{\sin(\theta)}\partial_\phi\psi^E(\mathbf{r},\omega)\end{aligned}$$

Therefore, we have (see **Page 428-429 of Jackson**) and noting that $\mathbf{e}_r\cdot\nabla\times\mathbf{g} = \frac{1}{r\sin(\theta)}\Big[\partial_\theta(g_\phi\sin(\theta)) - \partial_\phi(g_\theta)\Big]$:

$$-(1/k)\mathbf{e}_r \cdot \nabla \times (\mathbf{r} \times \nabla \psi^E(\mathbf{r},\omega)) = \\ -\frac{1}{kr\sin(\theta)}\left[\partial_\theta(\sin(\theta)\partial_\theta \psi^E(\mathbf{r},\omega)) + \frac{1}{\sin(\theta)}\partial_\phi^2 \psi^E(\mathbf{r},\omega)\right] = (L^2/kr)\psi^E(\mathbf{r},\omega) = \\ \Big[l(l+1)/kr\Big]\Psi^E(\mathbf{r},\omega),$$

where we also used that spherical harmonics are eigenstates of L^2. Therefore, we have:

$$\beta_F^2 \partial_r \rho(\mathbf{r},\omega) = (\omega_p^2 \varepsilon_0/k)(\partial_r \psi^L(\mathbf{r},\omega) + l(l+1)\psi^E(\mathbf{r},\omega)/r)$$

We also consider the requirement that the parallel component of the electric field remain unchanged across the sphere's boundary, which follows from Faraday's law. We first consider the $\mathbf{e}_\theta$ component (and noting that the $\mathbf{e}_\theta$ component of the curl for a vector field with no radial component is given by $\mathbf{e}_\theta \cdot \nabla \times \mathbf{g} = -(1/r)\partial_r(rg_\phi)$):

$$\mathbf{e}_\theta \cdot \mathbf{E}(\mathbf{r},\omega) = \frac{1}{k}\left[\frac{1}{r}\partial_\theta \Psi^L(\mathbf{r},\omega) + \frac{1}{r}\partial_r(r\partial_\theta \Psi^E(\mathbf{r},\omega))\right]$$

Therefore, if the spherical harmonics inside and outside of the sphere are the same, we need only worry about the continuity of the quantity $\Psi^L + \partial_r(r\Psi^E)$

We now consider the $\mathbf{e}_\phi$ component (noting that the $\mathbf{e}_\phi$ component of the curl of a vector field with no radial component is given by $\mathbf{e}_\phi \cdot \nabla \times \mathbf{g} = (1/r)\partial_r(rg_\theta)$):

$$\mathbf{e}_\phi \cdot \mathbf{E}(\mathbf{r},\omega) = \frac{1}{k}\left[\frac{1}{r\sin(\theta)}\partial_\phi \Psi^L(\mathbf{r},\omega) + \frac{1}{r}\partial_r\Big(r\frac{1}{\sin(\theta)}\partial_\phi \Psi^E(\mathbf{r},\omega)\Big)\right]$$

Therefore, we again only need to worry about the continuity of $\Psi^L + \partial_r(r\Psi^E)$ (that is, we don't obtain a boundary condition independent of the one already imposed on the $\mathbf{e}_\theta$ direction).

The last boundary condition results from requiring continuity of the parallel component of the magnetic field. This is equivalent to requiring continuity of $\nabla \times \mathbf{E}(\mathbf{r},\omega)$, or, equivalently, continuity of $\nabla \times \nabla \times \mathbf{L}\Psi^E(\mathbf{r},\omega)$ (with obviously no longitudinal contribution)

Note that $\nabla\cdot(\mathbf{r}\times\nabla\Psi^E(\mathbf{r},\omega))=\varepsilon_{ijk}\partial_i(r_j\partial_k\Psi^E(\mathbf{r},\omega))=0$. Therefore, we require continuity of $\nabla^2\mathbf{L}\Psi^E(\mathbf{r},\omega)=-k_M^2\mathbf{L}\Psi^E(\mathbf{r},\omega)$ (within the sphere) and $-k_0^2\mathbf{L}\Psi^E$ outside the sphere, which amounts to just requiring continuity of the radial component of the transverse component multiplied by the permittivity (since $\mathbf{L}$ only acts on the angular variables).

We now revisit the last boundary condition, which is imposed by the following

$$\beta_F^2\partial_r\rho(\mathbf{r},\omega)=(\omega_p^2\varepsilon_0/k)(\partial_r\psi^L(\mathbf{r},\omega)+l(l+1)\psi^E(\mathbf{r},\omega)/r)$$

Note that

$$\nabla\cdot(\nabla\Psi^L(\mathbf{r},\omega))=(k/\varepsilon_0\varepsilon_\infty)\rho(\mathbf{r},\omega)\rightarrow -k_{NL}^2\Psi^L(\mathbf{r},\omega)=(k/\varepsilon_0\varepsilon_\infty)\rho(\mathbf{r},\omega)$$

Therefore, we have that:

$$\left[\beta_F^2+(\omega_p^2/\varepsilon_\infty k_{NL}^2)\right]\partial_r\rho(\mathbf{r},\omega)=(\omega_p^2\varepsilon_0/k)l(l+1)\Psi^E(\mathbf{r},\omega)/r \text{ or equivalently}$$
$$\left[\beta_F^2+(\omega_p^2/\varepsilon_\infty k_{NL}^2)\right]\partial_r\Psi^L(\mathbf{r},\omega)=-(\omega_p^2/\varepsilon_\infty k_{NL}^2)l(l+1)\Psi^E(\mathbf{r},\omega)/r$$

By using our expression for k_{NL}^2, reiterated below:

$$k_{NL}^2=\frac{\omega_p^2}{\beta_F^2}\frac{1}{\varepsilon_\infty-\varepsilon_M}\frac{\varepsilon_M}{\varepsilon_\infty},$$

we may write the above as follows:

$$\beta_F^2(\varepsilon_\infty^2/\varepsilon_M)\partial_r\Psi^L(\mathbf{r},\omega)=-(\omega_p^2/k_{NL}^2)l(l+1)\Psi^E(\mathbf{r},\omega)/r$$

We now put everything together. We label the sphere radius with $\tilde{r}$. In the below, we don't include the factors of the spherical harmonics. In addition, we take $\Psi^L(\mathbf{r},\omega)=aj_l(k_{NL}r)$ for $r<\tilde{r}$ and null for $r>\tilde{r}$ (where there is no electron liquid). Similarly, we take $\Psi^E(\mathbf{r},\omega)=bj_l(k_Mr)$ and $j_l(k_0r)+h_l^{(1)}(k_0r)$ for $r<\tilde{r},r>\tilde{r}$, respectively. With this notation, we then have:

$$a j_l(k_{NL} r) + b\partial_r(r j_l(k_M r)) = \partial_r(r j_l(k_0 r)) + s\partial_r(r h_l^{(1)}(k_0 r))$$

$$b j_l(k_M r)\varepsilon_M = j_l(k_0 r) + s h_l^{(1)}(k_0 r)$$

$$a\frac{\varepsilon_\infty k_{NL}^2}{k}\frac{\beta_F^2 k_{NL}\varepsilon_\infty}{\varepsilon_M} j_l'(k_{NL} r) = -b\frac{\omega_p^2}{kr} l(l+1) j_l(k_M r) \text{ all evaluated at } r = \tilde{r}$$

Combining the first two equations, we have:

$$a j_l(x_{NL}) j_l(x_M)\varepsilon_M + (j_l(x_D) + s h_l^{(1)}(x_D))(x_M j_l(x_M))' = \left[(x_D j_l(x_D))' + s(x_D h_l^{(1)}(x_D))'\right] j_l(x_M)\varepsilon_M$$

Therefore, we have:

$$s\left[h_l^{(1)}(x_D)(x_M j_l(x_M))' - \varepsilon_M(x_D h_l^{(1)}(x_D))' j_l(x_M)\right] = \varepsilon_M j_l(x_M)(x_D j_l(x_D))' - j_l(x_D)(x_M j_l(x_M))' - a j_l(x_{NL}) j_l(x_M)\varepsilon_M$$

Now we need only substitute the value for a in order to solve for the scattered amplitude, s. For notational simplicity, we write $a = \alpha[j_l(x_M)/j_L'(x_{NL})]b$ and solve for α later.

From before, we know that $b = [j_l(x_D) + s h_l^1(x_D)]/j_l(x_M)\varepsilon_M$

Therefore, we have that

$a = \alpha[j_l(x_D) + s h_l^1(x_D)]/\varepsilon_M j_L'(x_{NL})$

Therefore, we have:

$$s\left[h_l^{(1)}(x_D)(x_M j_l(x_M))' - \varepsilon_M(x_D h_l^{(1)}(x_D))' j_l(x_M)\right] = \varepsilon_M j_l(x_M)(x_D j_l(x_D))' - j_l(x_D)(x_M j_l(x_M))' - \alpha[j_l(x_D) + s h_l^{(1)}(x_D)] j_l(x_{NL}) j_l(x_M)/j_L'(x_{NL})$$

So the contribution of nonlocality to the denominator of s will be $\alpha j_l(x_M) j_L(x_{NL}) h_l^{(1)}(x_D)/j_l'(x_{NL})$, and the full denominator of s will be given by:

$$\left[h_l^{(1)}(x_D)(x_M j_l(x_M))' - \varepsilon_M (x_D h_l^{(1)}(x_D))' j_l(x_M) + \alpha j_l(x_M) j_L(x_{NL}) h_l^{(1)}(x_D)/j_l'(x_{NL})\right]$$

Similarly, the contribution of nonlocality to the numerator of s will be $-\alpha j_l(x_M) j_L(x_{NL}) j_l(x_D)/j_l'(x_{NL})$ and the full numerator will be given by:

$$\left[\varepsilon_M j_l(x_M)(x_D j_l(x_D))' - j_l(x_D)(x_M j_l(x_M))' - \alpha j_l(x_M) j_L(x_{NL}) j_l(x_D)/j_l'(x_{NL})\right]$$

We now calculate α:

$$\alpha k_{NL}^2 \frac{\beta^2 \varepsilon_\infty}{\varepsilon_M} = -\frac{\omega_p^2}{x_{NL}\varepsilon_\infty} l(l+1)$$

We now substitute, $k_{NL}^2 = (\omega^2/\beta^2)\varepsilon_M/\varepsilon_\infty$, which gives us:

$$\begin{aligned}\alpha = -(\omega_p^2/\omega^2) l(l+1)/x_{NL}\varepsilon_\infty = (\varepsilon_M - \varepsilon_\infty) l(l+1)/x_{NL}\varepsilon_\infty \equiv \\ \Delta_{NL} j_l'(x_{NL})/j_l(x_{NL}) j_L(x_M),\end{aligned}$$

where the quantity Δ_{NL} is defined by **Equation 4c** of the Christensen paper. Therefore, we may write the quantity s as follows:

$$s = \frac{\varepsilon_M j_l(x_M)(x_D j_l(x_D))' - j_l(x_D)(x_M j_l(x_M))' - \Delta_{NL} j_l(x_D)}{h_l^{(1)}(x_D)(x_M j_l(x_M))' - \varepsilon_M (x_D h_l^{(1)}(x_D))' j_l(x_M) + \Delta_{NL} h_l^{(1)}(x_D)},$$

which is **Equation 4b** of the Christensen paper, as desired.